

Department of Computer Science**Assignment 3****Course Code: CS-151****Name:** _____**Course Title: Database Systems****Registration No.** _____**Total Marks: 20****Discipline/Semester: BSCS 2nd B****Date: 24-05-2025****Submission Date: 29-05-2025****Instructions:**

- **Late submissions will not be accepted.**
- **Copied assignments from any source (including AI) will be marked zero.**
- **Assignment must be hand written**

Q-1.**[CLO3, GA4, C4]**

The University of Wah, in collaboration with the NextGen Gaming Guild, is organizing a mega event titled **TECHON 2025**, scheduled to take place on the **2nd and 3rd of June 2025**. This national-level event includes a range of competitions in two major categories: **technical contests** and **gaming tournaments**. The event is hosted under the banner of the UW Computing Society.

The event lineup includes several technical competitions such as Speed Programming, Poster Designing, Project Exhibitions, Quiz Master (Technical), Escape Room: Tech Edition, and Thematic Videography. Alongside these, it also features gaming competitions like Tekken 8, FIFA 25, COD 4, and PUBG Mobile under the brand of “NextGen Gaming Battle Royal – Season 3”.

Each competition is scheduled at a specific venue and time, organized under different event teams, and has an assigned group of faculty coordinators and student volunteers. Volunteers serve in different operational capacities depending on the event requirements. The roles and responsibilities for these volunteers vary depending on the type of task assigned.

Students from within and outside the university can participate in one or more events. Registration is online, where participants must submit personal details and optionally request additional services such as certificates or branded merchandise. Certain competitions require solo participation while others require participants to form teams. For such events, a team leader handles the registration process and can invite others to join their team using a unique code.

Each event is judged by one or more individuals who are responsible for assessing participant or team performance. The judgment process and scoring criteria differ depending on the type of event. Scores are recorded, and winners are declared based on the event's evaluation method. Awards such as certificates, trophies, and cash prizes are distributed to the top performers.

If any event is canceled due to unforeseen circumstances, all associated data including schedules, participant registrations, teams, scores, judge allocations, and assigned volunteers must be removed from the system. The management system should support this data handling accordingly.

In some cases, the same person may serve multiple roles during the event or participate in different competitions, either solo or as part of different groups. Additionally, individuals working behind the scenes may hold different responsibilities even if they belong to the same organizational team.

Participants, judges, volunteers, and event records must be maintained in a digital event management system that allows registration, scheduling, team formation, score entry, and report generation. The system should be designed to support all necessary relationships, dependencies, data integrity, and reporting capabilities, while ensuring accurate representation of the entire TECHON 2025 structure.

Model a complete Enhanced Entity-Relationship Diagram using Crow's Foot notation. Your EERD must:

- Identify and represent all **strong and weak entities**.
- Use **appropriate relationships** with correct **modality and cardinality**.
- Include **specialization/generalization** (ISA hierarchy) wherever applicable in the scenario.
- Clearly model **aggregation and composition** structures.
- Represent **associative entities**, using both:
 - Composite primary keys
 - Surrogate keys (where identifier is assigned)
- Model **attributes properly**, including:
 - Simple (atomic)
 - Composite
 - Multivalued
- Apply **constraints** such as:
 - **Completeness constraints** (total or partial specialization)
 - **Disjointness/overlapping specialization**

Convert your EERD into a relational schema by applying the appropriate **mapping rules** discussed in the lectures (e.g., handling weak entities, multi-valued attributes, 1:N, M:N, and associative entities)

